

Cloud Fundamentals

1. Introduction to Cloud Computing

Cloud computing delivers computing services over the internet, including servers, storage, databases, networking, software, and analytics. It offers flexibility, scalability, and cost-efficiency by eliminating the need for physical infrastructure management.

2. Cloud Service Models

IaaS (Infrastructure as a Service): Provides virtualized computing over the internet (AWS EC2, Azure VMs).

PaaS (Platform as a Service): Offers development environment and tools (Google App Engine, Heroku).

SaaS (Software as a Service): Delivers applications via web (Salesforce, Microsoft 365, Slack).

3. Deployment Models

Public Cloud	Services available to the public over the internet
Private Cloud	Infrastructure used exclusively by one organization
Hybrid Cloud	Combination of public and private cloud resources
Multi-Cloud	Using services from multiple cloud providers

4. Major Cloud Providers

Provider	Market Share	Key Services
AWS	33%	EC2, S3, Lambda, RDS, SageMaker
Microsoft Azure	23%	VMs, App Service, Cosmos DB, AI Services
Google Cloud	11%	Compute Engine, Cloud Storage, BigQuery
IBM Cloud	7%	Hybrid cloud, OpenStack, AI services
Others	26%	Oracle, Alibaba, Digital Ocean, etc.

5. Cloud Architecture Principles

- Scalability: Easily add or remove resources based on demand
- Reliability: Redundancy and failover mechanisms ensure uptime
- Security: Multi-layered security, encryption, access controls
- Cost Efficiency: Pay-as-you-go model eliminates upfront costs
- Performance: Global infrastructure for low-latency access

6. Cloud Migration Strategies

- Rehost (Lift & Shift): Move applications with minimal changes
- Replatform: Make some optimizations for cloud
- Refactor/Re-architect: Redesign for cloud-native architecture
- Repurchase: Switch to cloud-based alternatives (SaaS)
- Retire: Decommission unnecessary systems

7. Security Best Practices

- Use strong authentication and MFA
- Implement least privilege access control
- Encrypt data in transit and at rest
- Regular backups and disaster recovery planning
- Monitor and audit cloud resources continuously